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Unsupervised learning for lithium-ion batteries fault diagnosis and thermal runaway early warning in real-world electric vehicles

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ABSTRACT

In this study, a multimodal diagnostic framework is developed to provide reliable early warnings of battery thermal runaway in real-world electric vehicles. Firstly, fault features of batteries are extracted from three dimensions: time, frequency and battery's internal resistance, preserving multidimensional fault information in both numerical and graphical forms. Secondly, a multimodal diagnostic algorithm for battery fault diagnosis is developed to provide early warning of thermal runaway to more than one week in advance through unsupervised learning with the joint inputs of multimodal data after graph feature extraction, dimensionality reduction and fault feature fusion. The algorithm enables the fusion and processing of fault data from different modalities, which enhances the ability to capture and analyze fault features. Thirdly, no complex pre-training or manually labeled data is required and the unified fault threshold allows for horizontal comparison. Finally, the proposed algorithm is validated using real-world data from different types of batteries in electric vehicles operating under various conditions. It shows excellent performances across different operating conditions and various battery types, achieving a balance of accuracy, reliability and generalizability.

1. Introduction

With increasingly stringent environmental requirements and growing emphasis on energy security globally, new electric vehicles (EVs) gradually replacing internal combustion engine vehicles [1]. Among these new energy vehicles, EVs are considered as the most promising option due to their advantages in environmental benefits, high performance, convenience, and intelligence. However, the complex and diverse operating conditions of EVs introduce significant safety risks to EV batteries, which is a major constraint to wide adoption of EVs. Battery faults can arise from electrical, thermal, and mechanical abuse [2,3], or production defects, leading to decreased battery capacity [4–6], limited power output, or even thermal runaway (TR) [7]. Therefore, accurate detection of battery faults and provision of early warning of TR are of great significance for promoting the development of EVs [8,9].

Battery fault diagnosis is developing rapidly in two directions. The first one is to apply new sensors such as mechanics and optical fiber, or the use of ultrasonic and impedance detection technologies to characterize the damage, deformation, pressure and temperature of the internal structure of a battery. The second one is to design advanced fault diagnosis algorithms based on existing electrical and thermal sensing signals [10,11]. Considering the cost and technical maturity, mainstream enterprises in the market mainly use the second technical route to detect battery faults. Existing fault diagnosis algorithms can be roughly divided into four categories: knowledge based methods, model based methods, signal processing based methods, and data driven methods [12]. Knowledge based methods, although widely used and well-established, struggle to build knowledge base and detect unknown faults [13]. Model based methods estimate fault parameters of mathematically modelled batteries, but it is challenging to fully simulate internal dynamics of batteries and balance between computational complexity and model accuracy. Signal processing based methods involve processing and analyzing signals collected by sensors during the operation of batteries to extract fault features, offering stable and robust fault detection [14]. However, signal decomposition methods are less

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effective in distinguishing progressive minor faults, and require further refinement to optimize decomposition parameters and fault diagnosis thresholds.

Data driven methods, which bypass the need for consideration of complex battery mechanisms, have emerged as a research hotspot in fault diagnosis [15]. Among them, machine learning techniques have strong capabilities in capturing dynamic characteristics of nonlinear battery systems, making them widely used in battery fault diagnosis. For instance, Wang et al. [16] adopted an improved radial basis function neural network to detect critical battery faults related to voltage, temperature and current. Yao et al. [17] used a wavelet based neural network to identify connection faults and employed a grid search support vector machine to detect potential abnormal states and classify fault severity. Zhang et al. [18] combined random forests, XGBoost and support vector machines to construct an ensemble learning model for diagnosing battery leakage faults. Zhao et al. [19] proposed a gated recurrent unit network combined with an ensemble multi-step prediction technique, effectively identifying battery overvoltage, undervoltage, overvoltage change rate, and cell consistency in a battery pack. Hong et al. [20] proposed a data partitioning method based on clustering and constructed a full-season coverage model using a spiral self-attention neural network to diagnose battery thermal abnormalities. Hong et al. [21] also developed a long short term memory neural network to perform precise multi-step voltage predictions, combined with alarm thresholds for fault diagnosis. Li et al. [22] explored a multi-step ahead thermal warning network based on core temperature detection for energy storage systems, providing early warning indications.

Although the machine learning methods are effective in many cases, they are still inadequate and require further refinement when they face complex real-world conditions in EVs. For example, it is difficulty to acquiring training data, evaluating data labels, and mapping single fault features to actual faults in real EVs. Additionally, integrating algorithms to handle multidimensional fault features, high computational demands and poor scalability hinders the practical application of fault diagnosis algorithms in real EVs. To address these issues, this study proposes an unsupervised multimodal fault diagnosis and warning framework for batteries based on real EV data [23]. By incorporating numerical and graphical features as inputs, the proposed framework expands the fault input modalities and increases the dimensions of fault features. It significantly enhances diverse fault information extraction and ensures compatibility with both sudden and gradual faults, achieving accurate diagnosis and localization of battery faults.

The main contributions of this study are as follows: (1) The proposed trend skewing measurement (TSM) feature effectively characterizes fault features over time scale, amplifying minor anomalies on a time scale. The short time Fourier transform (STFT) is introduced to obtain time-frequency spectrograms containing frequency domain information. Combining the modalities of frequency domain and time domain data enhances fault information and improves the ability to identify faults across multiple time scales. (2) A multimodal fault diagnosis algorithm is proposed for real-world battery fault diagnosis by constructing an unsupervised, multimodal and scalable framework. This framework can take various modal features such as numerical and image data as inputs, achieve automatic feature extraction and data dimensionality reduction for high-dimensional modalities without the need for complex parameter optimization and pre-training. It also realizes feature fusion and metric standardization for all the modalities, measures the degree of individual fault, and locates the faulty cell while maintaining strong scalability to accommodate future addition of new features. The algorithm is characterized by a well-defined logical structure, strong interpretability, and resistance to parameter degradation over time. (3) The accuracy, stability and ability to handle multimodal data of the proposed framework are validated using real-world EV data operating under normal, suspected abnormal and TR conditions.

The rest of the paper is organized as follows. Section 2 describes data and feature extraction methods. Section 3 introduces the multimodal fault diagnosis and warning framework. Section 4 presents the diagnostic performance of the framework using real-world EV data. Finally, Section 5 summarizes the key findings.

2. Data acquisition and feature extraction

2.1. Data acquisition and preprocessing

The data for this study is sourced from a Chinese auto company's big data monitoring platform. The dataset includes EVs that have caught fire, EVs with suspected TR, and normal EVs, complying with the Chinese national standard, GB/T 32960.1-2016. The average sampling rate of the data is 0.1 Hz.

Real-world cloud EV data, unlike laboratory data, often contains numerous anomalies and missing values and requires data preprocessing. The main types of the anomalies in the data include data disorder, voltage fluctuations, state of charge (SOC) fluctuations, etc. The data disorder, often due to data storage errors, can be corrected by rearranging based on mileage data. Voltage data abnormalities fall into two categories: data transmission faults and sensor faults. The former exhibit abnormal changes in data where all the cell voltages converge to a single point while the latter manifest as abrupt changes in single cell voltage value. The standard deviation of cell voltage is calculated for identifying voltage faults. Under normal circumstances, the cell voltages are randomly distributed within a certain range. When a transmission anomaly occurs, the cell voltages abruptly change to the same value, causing the standard deviation to drop significantly. Therefore, the ratio of the standard deviation of the voltage at a single sampling point to the average value of the standard deviation is used as the diagnostic criterion for transmission anomalies. The threshold of this ratio is set to 0.1 in this study to reduce the rate of missed diagnoses. Sensor anomalies are identified when cell voltage suddenly changes by $>$0.3 V and the difference between cell voltage and the average voltage of all the cells at that time is $>$0.2 V. Linear interpolation is performed to correct voltage anomalies using the mean of the data before and after the anomaly point. SOC anomalies, identified as jump $>$3 %, are also handled by linear interpolation.

After addressing the abnormal values, EV data segmentation is performed based on individual voltage and current data collected during each charge and discharge cycle. The national standard specifies EV state identification bits and charging status indicator bits as the basis for determining EV state. By segmenting the continuously collected data, we can obtain the corresponding charge and discharge data for each EV user.

2.2. Feature extraction

To comprehensively capture battery fault features, this study extracts the fault features from both intrinsic parameters and the dynamic response of a battery in terms of time domain and frequency domain, ensuring the completeness of fault information.

2.2.1. Charge resistance

Battery internal resistance strongly reflects the consistency and aging degree of a battery system, serving as an important basis for evaluating battery faults. However, the instability of voltage and current during EV operation poses challenges to determine internal resistance [24]. Fortunately, the multi-stage charging strategy commonly used in EVs cause voltage fluctuations in response to current switching, providing an opportunity for estimating internal resistance [25]. These voltage fluctuations during multi-stage charging process are illustrated in Fig. 1.

Although the estimated internal resistance derived from charging current switching does not precisely correspond to the actual ohmic resistance or polarization resistance, it still meets the requirements for cloud-based big data applications without strict adherence to physical accuracy [26,27]. The formula to calculate the internal resistance during

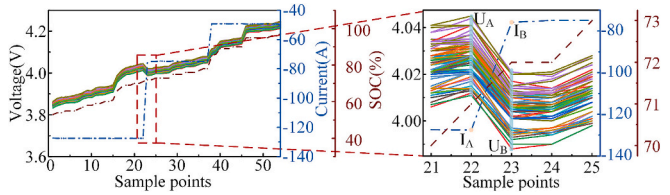


Fig. 1. Voltage fluctuations during multi-stage charging.

the current switching in the middle SOC range is shown as

$$Rc = -\frac{UA - UB}{IA - IB} \tag{1}$$

where R_c represents the charging internal resistance; U_A and U_B are the voltage just before and after current switching. I_A and I_B are battery current just before and after current switching.

2.2.2. Trend skewing measurement

During charge and discharge process, battery terminal voltage is inherently related to the SOC, charge and discharge rates, temperature, and internal state of the battery (e.g. internal resistance). To avoid the coupling effect between SOC and voltage on diagnostic algorithm, standard curves representing the normal voltage trend of each individual cell are established based on their respective voltage curves. The difference between the individual cell voltage and the standard voltage is defined as relative voltage (RV). A precise standard voltage curve is given by

$$V_N(i) = \frac{\sum_m^n V_x(i)}{m - n} \tag{2}$$

where $V_N(i)$ represents the standard voltage at time i, $V_x(i)$ represents the x-th cell voltage after sorting by voltage magnitude at time i. m and n mark the boundary points for the first 20 % and last 20 % of $V_x(i)$, respectively. This method excludes the top and bottom 20 % of the data to avoid the influence of outliers on overall voltage trend. With the standard voltage and individual cell voltage, the RV of each cell can be determined. The comparison between individual cell voltage and RV is shown in Fig. 2. Compared with individual cell voltage, RV decouples from SOC and emphasizes the differences in cell voltage when the current changes, which is more conducive to obtaining battery fault information.

Generally, the battery pack exhibits good consistency, with the voltage variations among the individual cells being largely uniform. As a result, the differences in RV between the individual cells are minimal.

Anomalies introduce slight voltage pattern variations between faulty and normal cells [28]. To highlight subtle voltage changes in faulty batteries, the TSM is proposed to accumulate the fault severity over time. The TSM integrates RV over time which is then averaged over the duration as follows

$$TSM_i = \frac{\sum V_{Ri}(t)}{t_{sum}} \tag{3}$$

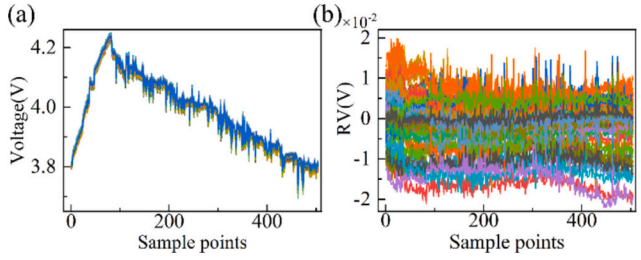


Fig. 2. Comparison between cell voltage and RV. (a) original cell voltage curves. (b) RV curves.

where TSM_i represents the TSM value of the i-th cell, $VR_i(t)$ represents the RV value of the i-th cell at time t, and t_{sum} represents the total duration to calculate TSM. This average value can avoid inconsistencies from varying lengths of discharge segments, facilitating a uniform threshold for fault detection and cell's TR risk determination. Considering that the cells prone to triggering TR often have internal short circuits, they manifest externally as slightly lower voltage compared to normal cells. To enhance fault diagnosis accuracy, especially for severe faults like internal short circuits, the TSM indicator is optimized as

$$TSM_i = \begin{cases} \alpha^* TSM_i (TSM_i > 0) \\ -TSM_i (TSM_i < 0) \end{cases} \tag{4}$$

where α is the scaling factor set to 0.8 to improve the diagnostic effectiveness for severe faults such as internal short circuits by reducing the weight of anomalies in higher voltage cells. The optimized TSM indicator, therefore, serves as a crucial input parameter for fault diagnosis.

2.2.3. Time-frequency diagram

Battery cell's response to a varying current contains critical fault information. Extracting this fault information from voltage curves via signal processing techniques, such as time-frequency transformations, can be highly effective. Fourier analysis, a common method to transform time domain data into frequency domain information, often struggles with non-stationary signals due to the loss of time domain information [29]. By converting a voltage signal into a RV signal and decoupling it from SOC, the signal exhibits stationary characteristics, providing a solid data foundation for Fourier transformation. One dimensional RV curves can be transformed into two dimensional time-frequency images using STFT. These images, containing both time domain and frequency domain information, enable the extraction of comprehensive information. Time-frequency images exhibit higher robustness to noise compared to time-series signals. In other fields, it has been demonstrated that time-frequency domain inputs outperform time domain inputs for fault diagnosis studies [30]. The schematic diagram of the STFT process is shown in Fig. 3.

Given a time domain signal $V(\tau)$, the STFT utilizes a window function $h(\tau)$ centered at time t to obtain a modified signal $V_t(\tau)$, which emphasizes the original signal $V(\tau)$ at time t while suppressing it at other times. In other words, non-stationary signals can be considered locally stationary and can then be transformed into the time-frequency domain [31,32].

$$V_t(\tau) = V(\tau)h(\tau - t) \tag{5}$$

Therefore, the modified signal is a function of both t and τ, given by

$$V_t(\tau) \simeq \begin{cases} V(\tau), \text{if } \tau \text{ is near to } t \\ 0, \text{if } \tau \text{ is far away from } t \end{cases} \tag{6}$$

The Fourier transform of the windowed signal reflects the frequency distribution around time t as

$$U_t(\omega) = \frac{1}{\sqrt{2\pi}} \int e^{-j\omega\tau} V(\tau)h(\tau - t)dt \tag{7}$$

The energy density spectrum at time t is

$$P_{SP}\left(t, \omega\right) = |U_t(\omega)|^2 = \left| \frac{1}{\sqrt{2\pi}} \int e^{-j\omega\tau} V(\tau)h(\tau - t)dt \right|^2 \tag{8}$$

For each time instance, the STFT generates a different spectrum. The sum of these spectra forms the spectrogram.

In converting the RV signal to a time-frequency image via STFT, two critical parameters must be set: the window width and the overlap width. A wider window width leads to higher frequency domain resolution, while a narrower window width leads to higher time domain resolution. According to the Heisenberg uncertainty principle, it's impossible to simultaneously achieve high resolution in both domains.

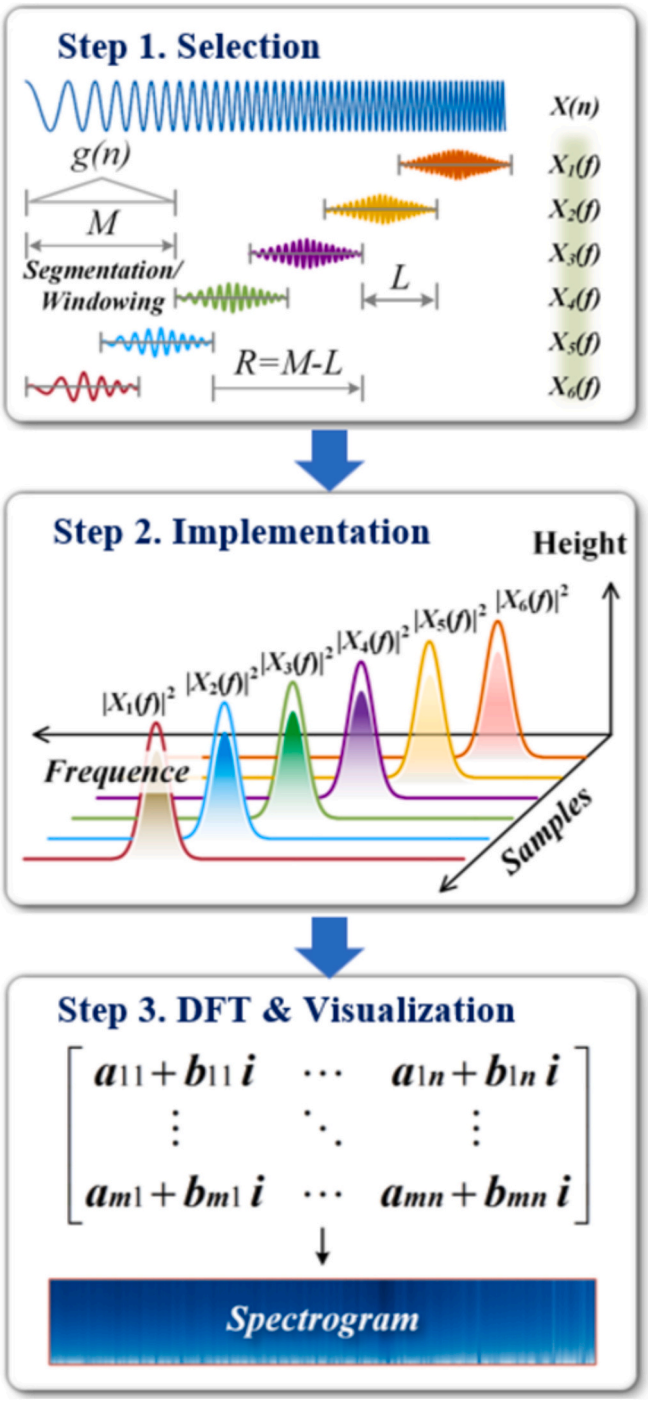


Fig. 3. Schematic diagram of the STFT process.

Reasonable time and frequency domain resolutions can make fault signals more prominent and reduce noise interference. After comparing the results at different combinations of the window width and the overlap width, we achieve the best results for fault feature extraction with the window width of 4 and the overlap width of 3.

The spectrogram obtained by STFT, stored as a time-frequency matrix S, contains both real and imaginary parts. For ease of subsequent processing, the matrix data needs to be transformed. The formula is

$$S_{edit} = 10^{*}log10(\mathrm{abs}(S)) \tag{9}$$

3. Multimodal fault diagnosis framework

To fully utilize the fault characteristics embedded in the time-frequency domain and achieve the fusion processing of multimodal fault features, we construct a unified diagnostic framework for integrating numerical and image multimodal inputs and fusion processing, realizing battery fault diagnosis. The proposed framework consists of a graph feature extraction module, a feature fusion processing module, and a multidimensional recurrent fault diagnosis module. The overall structure of this framework is illustrated in Fig. 4.

The constructed model is an unsupervised learning model, with inputs consisting of TSM (one-dimensional numerical features), internal resistance (one-dimensional numerical features), and time-frequency graph features (two-dimensional image features), as mentioned in Section 2. The framework automatically performs feature extraction and fault diagnosis, without the need for data labels or retraining.

3.1. Graph feature extraction module

This module extracts features from image data input and reduces the dimensionality of the features. Specifically, spectrogram data, which belong to the image modality, require complex feature extraction. This module consists of a convolutional neural network (CNN) including feature extraction layers and a kernel principal component analysis (KPCA) feature reduction layer. It performs primary feature extraction on the spectrogram data input in high dimension which are then fed into the KPCA layer for non-linear dimensionality reduction, achieving the transformation from high-dimensional image features to low-dimensional numerical features. After processing by this module, the original graph features are first transformed into 32-dimensional high-dimensional numerical features, which are then further reduced to one dimension, achieving an initial unification of the graph and numerical modalities.

3.1.1. CNN feature extraction layer

Considering the complexity of the image modality, CNN, which excels in image processing, is a natural choice. The overall framework of the CNN feature extraction network constructed in this section is illustrated in Fig. 4 (b). The core of the CNN is convolution and pooling. The convolutional layer extracts features from input data using a set of learnable convolutional filters or kernels. This process effectively captures local features of the input data and benefits from parameter sharing, reducing the number of model parameters.

The convolutional layer is a linear operation. To enhance the classification capability of the model, a non-linear activation function is typically applied after the convolution operation to introduce non-linear characteristics and increase the model expressive power. Commonly used activation functions include Sigmoid, Tanh, and ReLU, among others. However, due to the unique characteristics of the input time-frequency matrix, the commonly used ReLU function cannot effectively extract fault features. By replacing the activation function in the first layer with the ReLU6 function, successful feature extraction was achieved. Its formula is

$$f(x) = \begin{cases} 0, x \le 0 \\ x, 0 < x < 6 \\ 6, x \ge 6 \end{cases} \tag{10}$$

Compared to the ReLU function, ReLU6 limits the maximum activation to 6, providing better numerical stability and enhanced robustness for low-precision computations. It has a significant advantage over the ReLU function in convolutional layers when handling raw data inputs.

In the remaining layers, ReLU is used as the activation function with its formula being

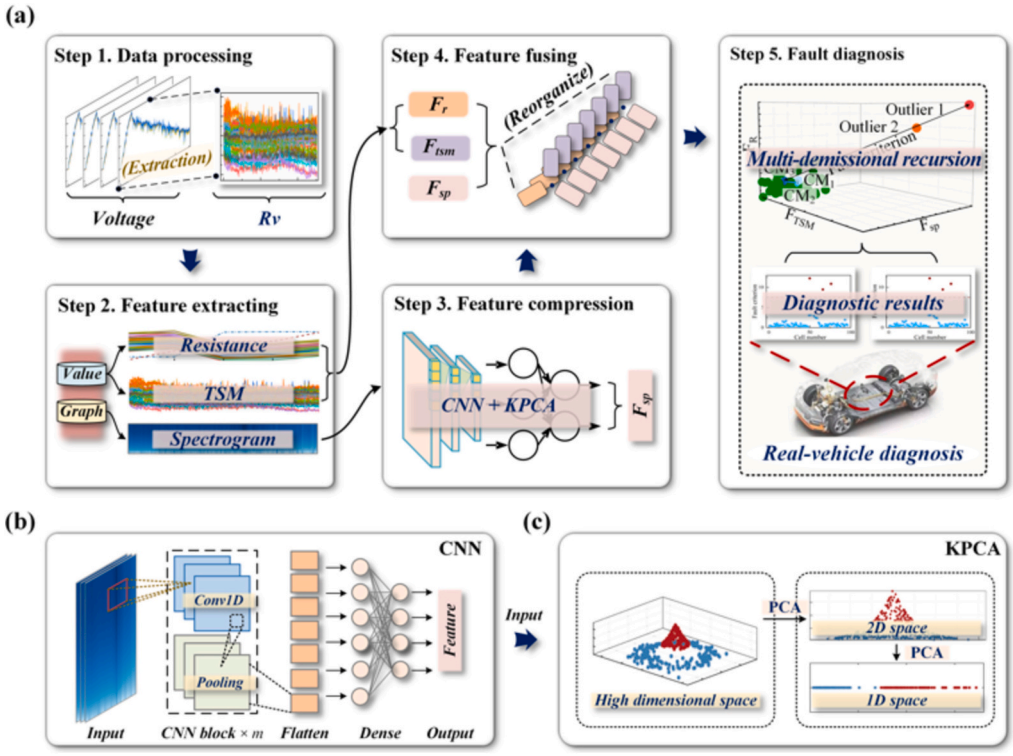


Fig. 4. Multimodal fault diagnosis framework. (a) overall flow chart; (b) structure of CNN graph feature extraction; (c) structure of KPCA feature dimensionality reduction module.

$$f(x) = \begin{cases} 0, x \le 0 \\ x, x > 0 \end{cases} \tag{11}$$

The advantages of the ReLU function lie in its low computational cost, non-linearity, ability to reduce model overfitting, and acceleration of model convergence. Additionally, it avoids truncating data for values >6, as in the case of ReLU6.

The pooling layer is used to reduce the spatial dimensions of feature maps, thereby reducing the number of model parameters, improving computational efficiency, and making the model more robust to changes in position.

Applying average pooling and max pooling methods, a fully connected layer is constructed to flatten and linearly transform the features extracted by the pooling layer into the multidimensional features. Since the input features are time-frequency spectrograms with uniformly distributed information, applying max pooling followed by average pooling to the input features shows better performance. Max pooling can better extract the most prominent parts of the feature map and determine the size and shape of the target. Average pooling helps reduce the size of the feature map and enhance the feature representation ability [33]. By combining average pooling with max pooling, effective extraction of fault features was achieved, reducing the issue of fault information being obscured by low-quality data. The main parameters of CNN are shown in Table 1.

By improving the activation function, selecting appropriate convolutional layer parameters, and designing the pooling layer rationally, the neural network we constructed successfully extracted fault features. In contrast, a standard CNN with the same number of layers was unable to extract abnormal features.

3.1.2. KPCA nonlinear dimensionality reduction

To avoid interference from high-dimensional features on other modal data in subsequent fault diagnosis, it is necessary to map high-dimensional features to low-dimensional space, achieving dimensionality reduction and uniformity in data dimension. The most common dimensionality reduction method is principal component analysis (PCA), which projects the original data into a new low-dimensional space through linear transformation to ensure that the projected data has the maximum variance, thereby preserving as much data information as possible. However, PCA has poor performance in dimensionality reduction when dealing with nonlinear data. The various feature dimensionality reduction and extraction methods we have attempted are shown in Table 2. It is necessary to introduce a new dimensionality reduction method. KPCA is a nonlinear extension of PCA, commonly used for dimensionality reduction of high-dimensional data and

Table 1
CNN layer structure and parameters.

Layer	Layer type	Kernel size	Function type
1	Convolution	4 * 4	/
2	Activation function	/	ReLU6
3	Convolution	3 * 3	/
4	Activation function	/	ReLU
5	Convolution	3 * 3	/
6	Activation function	/	ReLU
7	Convolution	3 * 3	/
8	Max pooling	3	/
9	Average pooling	/	/
10	Flatten	/	/
11	Linear	/	/

Table 2
Comparison of feature extraction method.

Extraction method	Result
Incremental component analysis (ICA)	Fail
Principal component analysis (PCA)	Fail
Spectral clustering	Fail
K-means	Fail
Sparse coder	Fail
KPCA	Success

discovering the main structures within the data. The basic idea of KPCA is to map the original data to a high-dimensional feature space using kernel techniques and perform PCA in that space. Compared to PCA, KPCA can handle nonlinear structures in the data and capture complex relationships among the data while preserving data information. The flowchart of KPCA process is shown in Fig. 4 (c).

The feature data extracted by CNN is denoted as $X = \{x_1, x_2, \ldots, x_n\}$, $x_i \in R_m$, $i = 1, 2, \ldots, n$. Here, m represents the dimensionality of the data, and R_m denotes the input space. To eliminate the influence of different scales in the sample data, the original data is first normalized. The normalization formula is

$$x_j = \frac{X_j - \min\limits_{1\le i\le n}\{X_i\}}{\max\limits_{1\le i\le n}\{X_i\} - \min\limits_{1\le i\le n}\{X_i\}} \tag{12}$$

where x_j is the normalized data, X_i is the original data of a certain feature vector, $min_{1\le i\le n}\{X_i\}$ and $max_{1\le i\le n}\{X_i\}$ are the maximum and minimum values of the feature vector, respectively.

Assuming φ is the mapping from the input space to the high-dimensional space, the feature space must satisfy

$$\sum_{i=1}^{n} \phi(x_i) = 0 \tag{13}$$

The covariance matrix is given by

$$C = \frac{1}{n}\sum_{i=1}^{n} \Phi(x_i)\Phi(x_i)^T \tag{14}$$

The eigenvalue λ of the matrix($\lambda \ge 0$) and eigenvector V should satisfy

$$\lambda V = CV \tag{15}$$

where the eigenvector of the covariance matrix is

$$V = \sum_{i=1}^{m} \alpha_j \phi(x_j) \tag{16}$$

The key of KPCA algorithm to achieve feature dimensionality reduction lies in the choice of kernel function. After comparing common kernel functions, we select Gaussian kernel function expressed by

$$K_{ij} = exp\left(-\frac{\left\|x_i - x_j\right\|^2}{\mathrm{N}}\right) \tag{17}$$

where K_{ij} is the feature similarity matrix, x_i and x_j are the i-th and j-th features, and N is the total number of features.

To simplify the eigenvalue polynomial of the covariance matrix, we use the kernel function as

$$n\lambda\alpha = K\alpha \tag{18}$$

In the KPCA algorithm, eigenvalues and eigenvectors are obtained from the kernel matrix. The cumulative contribution rate of the eigenvalues can be used to determine a number of the features extracted. Therefore, data dimensionality reduction can be achieved by selecting eigenvectors to form a new feature space. Since it is required to use the extracted features along with numerical features as inputs to the fault diagnosis module, the 32-dimensional feature data extracted by the CNN network is reduced to one dimension. In practical applications, the feature dimension can be chosen based on task requirements.

3.2. Feature fusion processing module

Significant differences among data modalities require uniform processing and fusion to achieve universality and comparability. This reduces algorithmic anomalies caused by varying modalities and ranges of the data. Since most cells in a battery module exhibit similar behavior, when unifying the measurements of data, normal cell feature values are chosen as the baseline and considered as one. Other feature values are then scaled proportionally. This process yields standardized features. Taking one set of TSM data as an example:

$$TSM_{normal} = \frac{\sum_m^n TSM(i)}{m - n} \tag{19}$$

where TSM_{normal} is the standard value of the TSM data set, $TSM(i)$ is the i-th cell voltage value sorted by TSM magnitude, and m and n are the boundaries of the top 20 % and bottom 20 % of the TSM data, respectively. Apply the following formula to each feature individually to obtain standardized features, which are then combined into a vector F_i to represent the three-dimensional features for each cell.

$$TSM_{uniform}(i) = \frac{TSM(i)}{TSM_{normal}} \tag{20}$$

After mean normalization and feature reorganization, the graph and numerical features are recombined to form new fault features, achieving the fusion of multi-modal data.

3.3. Fault diagnosis module

Considering that in a battery pack, anomalies and faulty cells are a minority and normal cells are the majority, it's possible to discriminate abnormal cells by finding the position of dense distributions of normal cells and assessing the degree of difference between abnormal and normal cells. The following is procedure to use the extracted three-dimensional fault features as input to identify faults in high-dimensional space.

Step 1, compute the centroid F_c of all the feature vectors in the three-dimensional space:

$$\begin{cases} F_{xc} = \dfrac{1}{n}\sum\limits_{i=1}^{n} F_{xi} \\ F_{yc} = \dfrac{1}{n}\sum\limits_{i=1}^{n} F_{yi} \\ F_{zc} = \dfrac{1}{n}\sum\limits_{i=1}^{n} F_{zi} \end{cases} \tag{21}$$

where F_{xi}, F_{yi}, F_{zi} represent three dimensions of the feature vector, F_i is the number of feature vectors.

Step 2, calculate the Euclidean distance between each cell's feature vector and the centroid:

$$D_i = \sqrt{(F_{xi} - F_{xc})^2 + (F_{yi} - F_{yc})^2 + (F_{zi} - F_{zc})^2} \tag{22}$$

Step 3, calculate the mean of all the cell Euclidean distances:

$$D_{pre} = \frac{1}{n}\sum_{i=1}^{n} D_i \tag{23}$$

Step 4, calculate the mean distance after removing the maximum value from D_i:

$$D_{aft} = \frac{1}{n\text{-}1}\sum_{i=1}^{n\text{-}1} D_i \tag{24}$$

Step 5, calculate the ratio of D_{aft} to D_{pre}:

$$\beta = \frac{D_{aft}}{D_{pre}} \tag{25}$$

With $\beta \ge 0.95$ as the threshold, we perform multiple iterations until the threshold is reached. Although abnormal cells in a battery pack have relatively low proportion, their differences from normal cells are significant, which tend to contribute larger distances to the mean distance distribution. By iteratively removing abnormal cells, interference from

abnormal cells in the mean calculation can be reduced, and a data centroid position that better represents normal data can be obtained.

Step 6, calculate the final distance D_{fi} of each cell to the final centroid position:

$$D_{fi} = \sqrt{\left(F_{xi} - F_{xfc}\right)^2 + \left(F_{yi} - F_{yfc}\right)^2 + \left(F_{zi} - F_{zfc}\right)^2} \quad (26)$$

where F_{xfc}, F_{yfc} and F_{zfc} represent the final centroid position.

Step 7, determine the occurrence of a fault based on the fault threshold. Through the analysis of the principle of this method and summarizing the actual application results on multiple vehicles' data, setting the fault threshold at $D_{fault} = 7.5$. Exceeding this threshold is considered indicative of a fault. Verification has shown that this threshold effectively balances the false positive rate of normal cells and the false negative rate of faulty cells.

Once the threshold is determined, the model is fully constructed. During use, simply input the new data into the framework to automatically identify faults without the need for pre-training.

4. Results and discussions

4.1. Diagnostic results

To verify the effectiveness of the algorithm, we will analyze a large dataset from nine vehicles, assessing the algorithm's performance across different vehicles, operating conditions, and battery systems. Detailed analyses will focus on one vehicle that caught fire, one vehicle suspected of TR, and one normal vehicle. The cell voltage curves of the three vehicles are shown in Fig. 5.

4.1.1. Results for TR vehicle

The vehicle analyzed in this study experienced TR in mid-April 2023. Cloud monitoring data from the manufacturer was obtained for 14 days preceding the TR event. The vehicle, a passenger car, was equipped with a battery pack consisting of 96 cells organized into 8 modules, each containing 12 cells. The cells were connected in a 4-series, 3-parallel configuration. Detailed parameters of the individual battery cells are shown in Table 3. The vehicle traveled a total mileage of approximately 200,000 km.

Within the data range, the vehicle traveled a total of 3228 km and underwent 20 charging events. The cell voltage ranged from 3.26 to 4.25 V, the current ranged from −176.2 A to 234.5 A, with an average discharge current of 19.2 A. The minimum SOC was 18 %, and the battery temperature ranged from 15 to 37 °C, all within the normal operating range, with no abuse events occurring.

The manufacturer retrieved and disassembled the TR vehicle. The investigation confirmed that cell #76 triggered the TR, which corresponded to the sudden voltage drop observed in cell #76 prior to the incident. Additionally, the disassembly revealed that cells #50 and #66 also exhibited faulty behavior.

Table 3
Cell parameters.

Parameter	Value
Battery type	Ternary
Size	148 mm × 27 mm × 92 mm
Weight	0.85 kg
Rated capacity	50 Ah
Working voltage	3.65 V
Charging cut-off voltage	4.25 V
Discharging cut-off voltage	3 V

According to the recorded data from April 1 to April 14, 2023, the voltage of cell #76 dropped sharply on the day of the incident, which occurred approximately 30 min after switching from charging to driving, then about 10 s later this vehicle experienced TR and the battery management system in the vehicle stopped recording data. The algorithm proposed in this study provides fault diagnosis results for the dataset as shown in Fig. 6. Due to the algorithm using discharge segments, there are two sets of discharge segments on April 4th, 7th, and 10th, as the vehicle was recharged and continued to be used after the initial discharge. There are no diagnostic results for April 13th, as there was no driving data available for that day. Prior to the occurrence of TR on April 14, the fault score of cell #76 was relatively high but had not yet exceeded the threshold before April 4. However, the fault scores of cells #50 and #66 had surpassed the threshold with most of the subsequent time spent above it. This aligns with the result drawn from the disassembly that cells #50 and #66 also had faults. On April 4, the fault score of cell #76 exceeded the threshold for the first time, marking the algorithm's initial detection of the fault, which persisted until the occurrence of TR. The abnormality level of cells #49–#60 was slightly higher than those of normal cells but did not surpass the threshold. This could be attributed to the slight differences in the parallel-connected cells within the same module caused by the abnormality in cell #50 [34,35].

The diagnostic results for the vehicle indicate that the proposed method can capture battery anomalies for a relatively long period (>7 days) before TR. Thanks to the cumulative effect of abnormal TSM features over time, the proposed method demonstrates good stability in fault diagnosis, enabling stable fault detection. Benefiting from metric normalization, the proposed algorithm achieves accurate differentiation of abnormal cells from normal cells with their fault scores consistently below the threshold. If we exclude cells #49–#60, which is influenced by cell #50, the fault scores of the remaining normal cells are all significantly lower than the threshold, greatly reducing the risk of

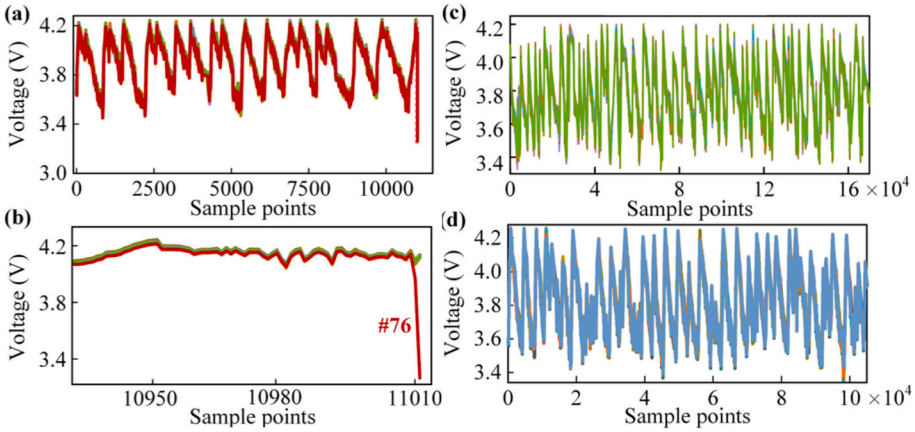


Fig. 5. Cell voltages (a) TR vehicle; (b) enlarged view of the TR vehicle before the occurrence of TR; (c) suspected TR vehicle; (d) normal vehicle.

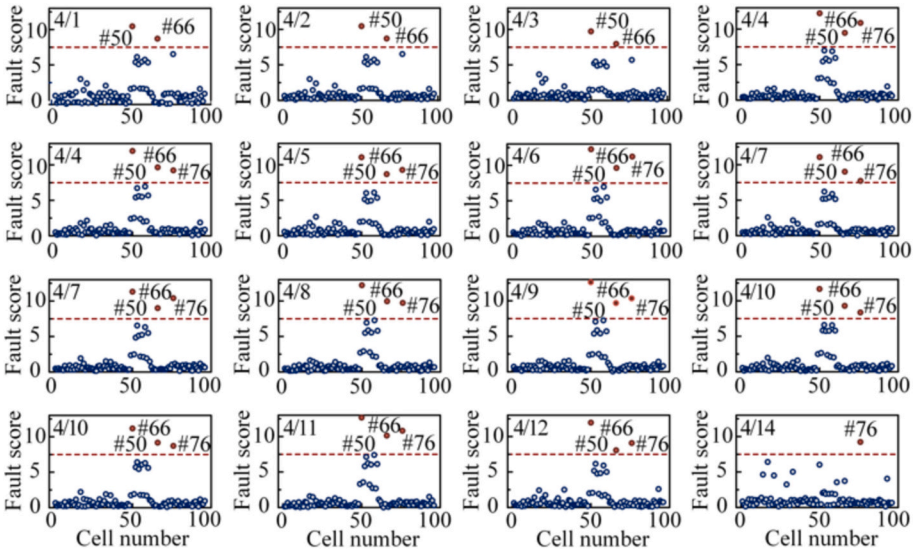


Fig. 6. TR vehicle fault diagnosis results.

misjudgment.

In addition, we obtained three other TR vehicles equipped with lithium iron phosphate battery modules from our partners. Due to space constraints, the results and analysis of these vehicles will be presented in the Supporting information. The proposed algorithm successfully issued thermal runaway warnings 11, 10, and 8 days before the events occurred. Overall, the proposed algorithm demonstrates reproducibility in identifying vehicles at risk of thermal runaway, with an average warning time of approximately 11 days, thereby achieving a week-level thermal runaway warning.

4.1.2. Results for suspected TR vehicle

To further validate the effectiveness of the proposed algorithm, we obtained a dataset from our partners containing data for an extended period before TR occurrence. The vehicle's topology and battery type are identical to those of the previously mentioned vehicles. The data were collected from February 10, 2021, to May 10, 2021, with the vehicle covering a mileage range from 93,568.5 km to 101,247.3 km, with a current range of −117.3 to 341.5 A, an average discharge current of 14.48 A, voltage ranging from 3.32 to 4.20 V, a minimum SOC as low as 2 %, and a battery temperature range of 9 to 38 °C, all within the normal operating range. The cell voltage data for this vehicle are displayed in Fig. 5 (c). It is almost impossible to distinguish abnormal cells from normal cells.

Compared to the previously mentioned TR vehicles, this dataset records operational data for an extended period leading up to TR. As of the data cutoff point, the vehicle has not experienced TR. However, several days later the vehicle caught fire. This implies that, as of the data cutoff point, the vehicle remained in a suspected TR phase. According to the disassembly conducted by our partners, cell #41 was identified as the origin of TR, with cells #9 and #93 also exhibiting faults. TR subsequently propagated to the surrounding cells after the initial TR, ultimately leading the vehicle to catching fire. Fig. 7 (a) shows the diagnosis results of the suspected TR vehicle.

It can be seen that the fault scores of all the individual cells were below the threshold at the beginning of data collection, then cells #9, #41, and #93 exhibited relatively high anomaly scores. Over time, on February 27, March 16, and March 31, the respective fault scores did not exceed the threshold, but their fault scores gradually increased. It wasn't until April 17 that the fault score of cell #41 first surpassed the threshold. On May 10, the fault scores of cells #9 and #93 also exceeded the threshold. Several days later, the vehicle experienced TR. Due to the lack of data beyond this point, we cannot ascertain the algorithm's performance closer to the TR event. Nevertheless, this data indicates that the proposed algorithm demonstrates good diagnostic efficacy for cells with suspected TR risks.

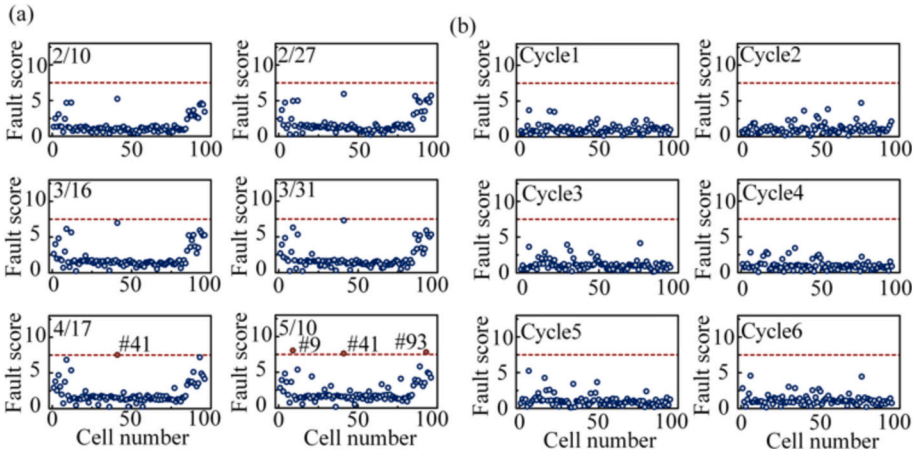


Fig. 7. Diagnosis results (a) suspected TR vehicle (b) normal vehicle.

4.1.3. Results for normal vehicle

To further validate the reliability of the proposed algorithm, we selected the data from several normal vehicles to assess the algorithm's performance on normal data. Here, we present an example of one such normal vehicle. The recorded data for this selected vehicle spans from January 1, 2022 to March 15, 2022. As of now, this vehicle has not experienced any faults. The vehicle utilizes a topology with large capacity ternary lithium battery cells connected in series.

The fault diagnosis results for this vehicle are shown in Fig. 7 (b). Throughout the recorded data period, the fault scores remained consistently below the fault threshold. The fault scores of all individual cells were maintained at relatively low levels. There was no instance of any cell exhibiting high anomaly scores, nor did any individual cell's fault score exhibit a continuous increase over time. Results for the other normal vehicles were similar to those of the vehicle presented, demonstrating the high reliability and low false positive rate of the proposed algorithm.

4.2. Discussions

4.2.1. Evaluation of feature effectiveness

This method proposes three types of fault feature extraction methods. Among them, the numerical feature TSM primarily demonstrates the accumulation of faults over time, which are mainly used to identify progressive faults. The charge internal resistance mainly reflects the internal state of the battery and plays an auxiliary role in fault assessment. The spectral feature mainly represents dynamic fault characteristics of a battery, with better diagnostic effectiveness for sudden occurrence of faults. To validate the effectiveness of the proposed features, this study analyzes the relevant features using faulty vehicles as examples.

4.2.1.1. Evaluation of TSM feature effectiveness. In the process of obtaining TSM features, outliers are first removed from the individual cell voltage curves to accurately extract the standard voltage curve. This reduces the interference of anomalous values on the process of extracting the standard curve. Taking the discharge data before TR as an example, we compare the median, mean, and RV curves in Fig. 8.

It can be observed that the median curve is influenced by some anomalous cells, resulting in a lower median curve compared to the mean curve and RV curve at most sampling points. Additionally, since the median curve cannot reflect the overall trend of the data, it may fail to capture the distribution trend of normal cells when the data distribution is dense but contains outliers. Compared to the median curve, the mean curve better reflects the overall trend of the voltage curve. However, it is most affected by outliers, as evidenced by the sharp drop in voltage at the last moment of data for cell #76 as shown in Fig. 5 (b).

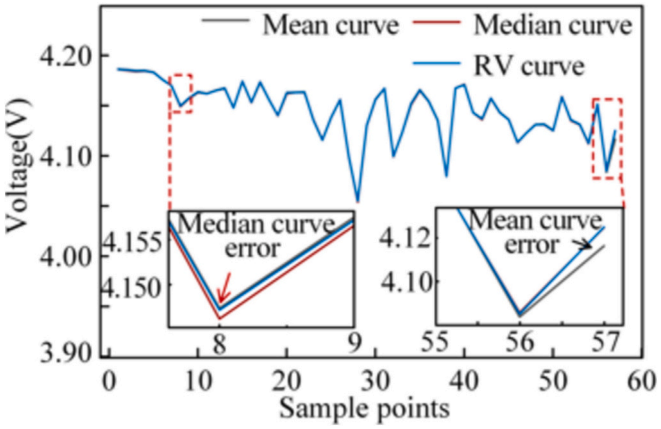


Fig. 8. Comparison of mean, median, and RV curves.

This indicates that the mean curve has a lower voltage compared to other curves, being the most affected by outliers. In summary, the RV curve used can better reflect the trend of normal cells and avoid potential biases caused by improper selection of voltage reference values for subsequent algorithms.

Using the data from the same faulty vehicle as an example, we will verify the effectiveness of the TSM. From Fig. 9, it is evident that TSM successfully identifies anomalous cells through the accumulation of voltage anomalies. However, the differentiation between faulty cells and normal cells is relatively poor because the distinction between them is often minor before TR occurs. It is challenging to use a single feature to set a unified threshold for faulty cell isolation despite being able to distinguish anomalous cells from normal cells. Hence, this study introduces frequency domain features as a complement to time-scale features.

4.2.1.2. Evaluation of frequency domain feature effectiveness. STFT can extract frequency domain features from RV and generate frequency domain information in the form of spectrograms. Still using the data from the faulty vehicle as an example, we display a partial time-frequency map of RV after STFT on April 1 in Fig. 10.

Cells #1, #45, and #95 represent normal cells, while Cells #50, #66, and #76 represent faulty cells. The colors in the spectrogram represent the values of the time-frequency matrix after the STFT transformation. Compared to abnormal cells, normal cells exhibit more pronounced fluctuations on the time scale and show distinct differences in their distribution on the frequency scale. The colors also display significant variation, indicating that the time-frequency matrices of normal and abnormal cells are markedly different.

Considering the randomness of real-world data, we attribute these fluctuations to abrupt change of the components in the signal causing significant changes in the frequency components at different time points. To distinguish these differences, we need to employ CNN-KPCA for joint feature extraction and dimensionality reduction of spectrograms.

Fig. 11 show the feature extraction of time-frequency maps using CNN-KPCA. The anomalous feature values of the three TR cells are all relatively high. However, notably the cells within the same module as cell #50 also exhibit high fault feature values, which were not evident in the TSM features. This suggests that the extraction of frequency domain information by STFT makes the anomalous response of individual cells more pronounced. The anomaly in cell #50 results in abnormal responses in the entire module, indirectly confirming the earlier hypothesis that the anomaly in one cell can lead to abnormal behavior in adjacent cells. However, while the obtained features successfully differentiate TR cells from normal cells, they may still result in a relatively high false positive rate. Therefore, it is necessary to comprehensively apply multidimensional features to diagnose faults, which improves fault detection accuracy and reduce false positive rates.

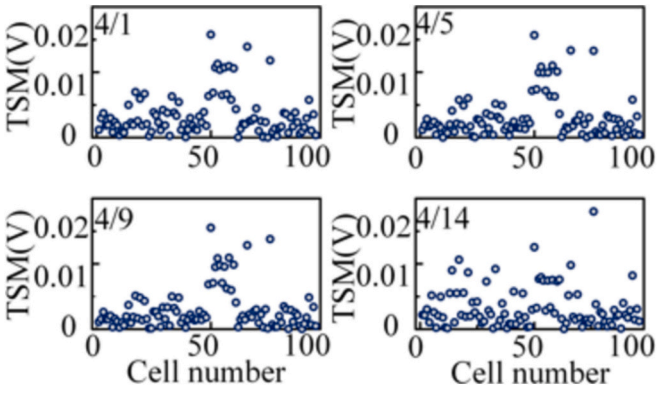


Fig. 9. TSM for the TR vehicle.

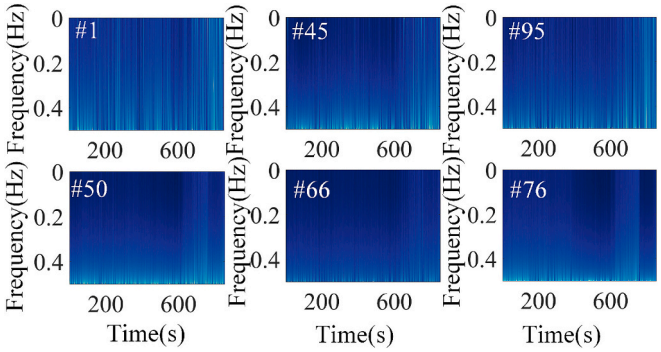


Fig. 10. Spectrograms of cell RV for a normal vehicle (above) and a TR vehicle (below).

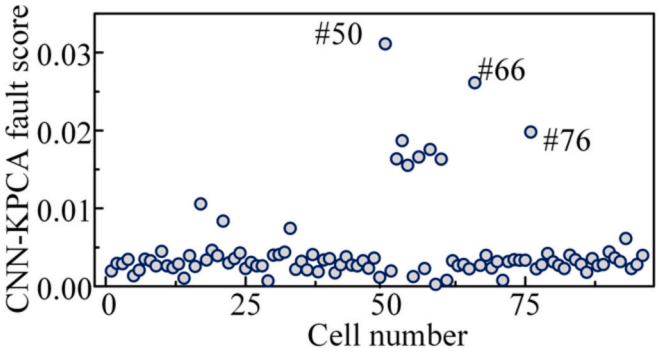


Fig. 11. Results of CNN-KPCA method.

4.2.2. *Evaluation of anomaly discrimination method effectiveness*

4.2.2.1. Evaluation of metric uniformity method. To ensure that three types of features extracted in this study have the same weight when inputting into the fault diagnosis algorithm, it is necessary to normalize the extracted features. Due to the clustering characteristics of battery data, where features of normal cells are expected to exhibit a dense distribution while abnormal cells show an outlier distribution, conventional normalization and standardization methods can be affected by these outliers, making them unsuitable for such data. Therefore, we use the mean value of normal cells as a reference, scaling it linearly to 1, and then determining the magnitude of other data based on this reference. This method, compared to conventional approaches, is less affected by outliers and is more suitable for the actual conditions.

The TR vehicle is still used as an example to compare the results before and after normalization in Fig. 12. For normal cells, the range of TSM features is 0–0.016 V with a mean value of 0.0026 V before normalization. The range of graph features is 0–0.31 with a mean value of 0.0031. The range of charge internal resistance is 0–1.7 * 10^{-4} Ω with a mean value of 9.8 * 10^{-5} Ω. After normalization, the mean values of the input features are all 1, significantly reduce errors caused by large differences in the data ranges of different features.

Compared to conventional standardization algorithms, the proposed method performs normalization based on the characteristics of real vehicle battery packs, avoiding normalization anomalies caused by different degrees of outliers in each feature.

4.2.2.2. Evaluation of multidimensional recursive fault diagnosis effectiveness. After obtaining feature values with unified metric scales, we apply the multidimensional recursive fault diagnosis module to iteratively eliminate outliers, recalculate centroids, and gradually find the centroid position that best represents normal cells.

Using data from a malfunctioning vehicle on a specific day as an example, we applied a multidimensional recursive algorithm to calculate the centroid positions before and after recursion, as well as the distance from each cell to the centroid. We found that the average distance from normal cells to the centroid decreased from 1.0766 to 0.6488, a reduction of 39.7 %. The reduction in distance indicates that after removing the abnormal cells, the centroid position is closer to the centroid of the normal cells, effectively reducing the probability of misjudging normal cells during the fault diagnosis process. The diagnostic results for three abnormal cells improved from 11.19, 7.41, and 8.41 to 11.94, 8.06, and 9.07, respectively, with an average improvement in discriminative ability of 7.56 %. It has to some extent improved the fault diagnosis algorithm's ability to distinguish faulty batteries. When analyzing other segments, we found that the overall average discriminative ability for defective cells improved by 7.34 %. The above results indicate that the multi-level recursive algorithm enhanced the ability to identify abnormal cells during the recursion process, reduced the fault scores of normal cells, and contributed to improving the accuracy of fault diagnosis. This plays a significant role in reducing both the false negative and false positive rates. A comparison of the results before and after processing is shown in Fig. 13.

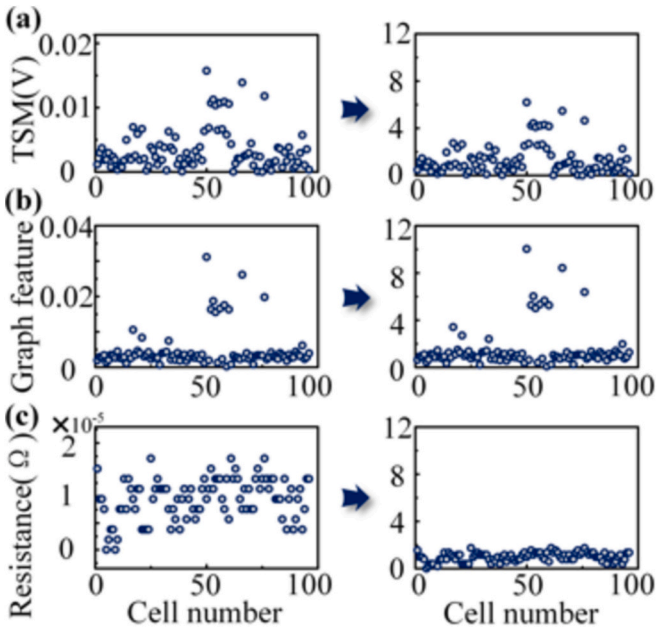


Fig. 12. Metric uniformity. (a) TSM; (b) Graph feature; (c) Resistance.

4.3. *Evaluation of algorithm universality*

To further validate the algorithm's universality across different

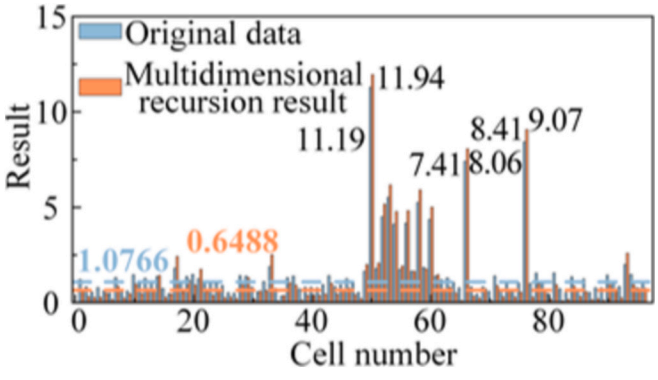


Fig. 13. Multidimensional recursion result.

operating conditions and battery systems, we select the data from city buses equipped with lithium iron phosphate (LFP) batteries for verification. Among these buses, one experienced TR while another operated normally. This bus that experienced TR has a total of 378 cells, connected in a 2-parallel 189-series configuration. Detailed parameters are shown in Table 4. The partner provided data only for the last cycle before the occurrence of TR event. The voltage data for individual cells in this cycle are shown in Fig. 14.

TR occurred near the 400th sampling point, starting with cell #372. Subsequently, the other cells in the battery pack experienced TR under the influence of cell #372. This is verified by the disassembly results in which there was the indications of a fault in cell #372 and no obvious faults in the other cells. Hence, it can be inferred that the subsequent TR of the other cells was induced by the TR of cell #372. In the following section, the data segments before the occurrence of faults, specifically the data before the 400th sampling point, were selected to demonstrate the algorithm's diagnostic capabilities in the absence of obvious voltage anomalies.

The fault diagnosis results are shown in Fig. 15. Only the fault score for cell #372 exceeds the threshold, while the fault scores for the other cells are significantly below the threshold. This aligns with the individual cell voltage curves and disassembly analysis results, confirming that the algorithm can accurately diagnose faults for LFP batteries under urban bus road conditions.

Unlike the data from the TR vehicle, the data for the normally operating vehicle was provided by the partner. This vehicle consists of 162 battery cells, with the same cell parameters as those in Table 4. No battery cell failures occurred during the operating period of this vehicle. The diagnosis results for this vehicle are shown in Fig. 16.

As the partner provided data for a certain period of time, here we only present the data for 8 cycles. The results for the remaining cycles are consistent with the displayed data. For this vehicle, the fault score for each cell is relatively low, with a significant distance from the threshold. Regarding the issue where cells with larger serial numbers have higher fault scores compared to those with smaller serial numbers, we realized through discussions with our partners that this is due to differences in battery connection methods, which result in varying responses between the first half and the second half of the battery cells. Therefore, although the fault scores in the second half are relatively higher, they remain within the normal range, indicating that the algorithm has redundancy for normal data and a low false alarm rate.

5. Conclusions

This study proposes a real EV battery fault diagnosis and TR early warning framework based on multimodal method. The algorithm leverages time-scale TSM and battery charge resistance as numerical fault features and spectrograms as graph features. Through CNN-KPCA, graph features are combined and reduced to lower-dimensional features, forming a multidimensional recursive fault diagnosis module for early battery fault diagnosis and TR warning. The main conclusions are as follow:

(1) TSM features enhance fault characteristics over time-scale, facilitating the detection of gradual and minor faults. Time-frequency graph features represent fault characteristics in the

Table 4
LFP cell parameters.

Parameter	Value
Battery type	LFP
Rated capacity	302 Ah
Working voltage	3.22 V
Charging cut-off voltage	3.5 V
Discharging cut-off voltage	2.65 V

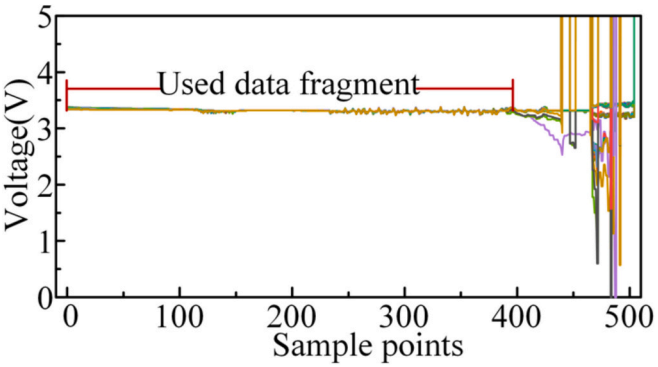


Fig. 14. Cell voltages of LFP batteries in TR vehicle.

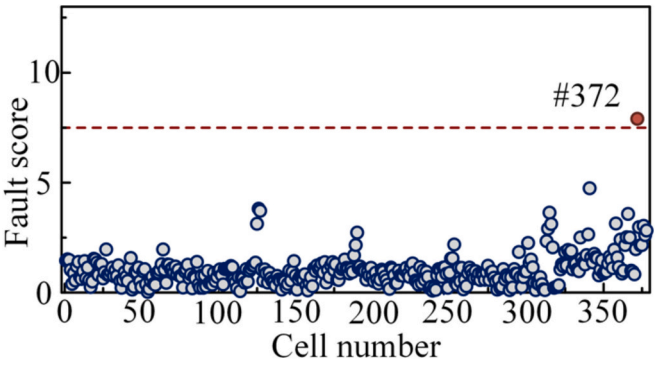


Fig. 15. Fault diagnosis results.

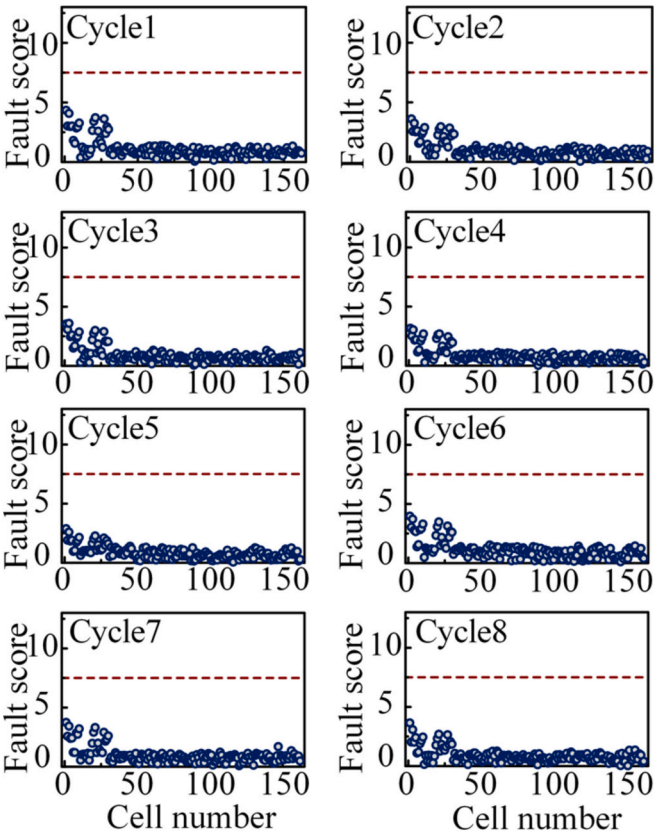


Fig. 16. Diagnosis results of LFP batteries in normal vehicle.

frequency domain, highlighting sudden occurrence of faults. Charging resistance reveals internal abnormalities to identify internal short-circuit faults. These features have strong fault discrimination and complement each other.

(2) The multimodal diagnosis algorithm stably and accurately diagnoses faults by jointly evaluating various data types. It achieves fault diagnosis by comparing differences between cells, thereby avoiding inaccuracies caused by battery aging. The diagnostic results for multiple TR vehicles indicate that the algorithm achieved an average TR warning time of 11 days. The unsupervised learning approach reduces computational complexity and costs, eliminating the need for large-scale data labeling and model pretraining. Additionally, the algorithm's scalability allows for enhanced feature dimensions without retraining. Compared to algorithms that require retraining, the workload can be reduced by up to 50 % [36].

(3) Validation on two TR vehicles, one suspected vehicle, and three normal vehicles showed the proposed algorithm can detect faults and provide warnings at least 10 days in advance, with minimal false alarms for normal vehicles. The algorithm performs well across different battery chemistries (LFP and ternary lithium batteries) and operating conditions (passenger cars and city buses), indicating strong generalizability.

Further study is needed to enhance feature screening for additional modal inputs. It is also needed to study the prediction of the remaining time before TR due to the random nature of chronic TR and varying driver behavior.

CRediT authorship contribution statement

Quanqing Yu: Writing – review & editing, Writing – original draft, Methodology. **Yu Yang:** Writing – original draft, Methodology, Investigation. **Aihua Tang:** Writing – review & editing, Investigation, Data curation. **Zikang Wu:** Investigation, Data curation. **Yuchen Xu:** Visualization, Validation, Data curation. **Weixiang Shen:** Writing – review & editing. **Feng Zhou:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.est.2024.115194.

Data availability

The data that has been used is confidential.

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